Chapter 2 Answers

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Lemma

We must show that, from F_1 and F_2, the following two conditions hold:
(1): We need to show that $\sim\exists x\phi(x)$ is true iff $\sim\phi(a)$ is true for every parameter a. Suppose that $\sim\exists x\phi(x)$ is true (under some valuation v). Thus, by F_2, the sentence $\phi(a)$ is not true under v for any parameter a. Because $\phi(a)$ is not true under v for any parameter a, it follows that $\sim\phi(a)$ is true under v for every parameter a (by B_1). On the other hand, suppose that $\sim\phi(a)$ is true under v for every parameter a. Thus, by B_1, $\phi(a)$ is false under v for every parameter a (by B_1). Thus, by F_2, $\exists x\phi(x)$ is not true under v, meaning, by B_1, $\sim\exists x\phi(x)$ is true under v.
(2): We need to show that $\sim\forall x\phi(x)$ is true iff $\sim\phi(a)$ is true for at least one parameter a. On the one hand, suppose that $\sim\forall x\phi(x)$ is true under some valuation v. Thus $\forall x\phi(x)$ is not true under v (by B_1), meaning, by F_1, $\phi(a)$ is not true for every parameter a. Thus, for the parameter e for which $\phi(e)$ is not true, $\sim\phi(e)$ is true by B_1. On the other hand, suppose that $\sim\phi(a)$ is true for at least one parameter a, call it e. Thus, by B_1, $\phi(e)$ is false, meaning, by F_1, $\forall x\phi(x)$ is not true, and therefore, by B_1, $\sim\forall x\phi(x)$ is true.

1

We suppose that v is a Boolean valuation such that the following two conditions hold (for all sentences γ and δ):
(1) if γ is true under v, then for every parameter a, the sentence $\gamma(a)$ is true under v, and
(2) if δ is true under v, then $\delta(a)$ is true under v for at least one parameter a.
We must prove that v is a first-order valuation. Thus, we must derive F_1' and F_2'. Because we have already have (1) and (2), we only need to prove the converses

of (1) and (2).
(1): In this case, we need to prove that, if for every parameter a, the sentence $\gamma(a)$ is true under v, then γ is true under v.

Case 1: In this case, γ is of the form $\forall x\phi(x)$. So suppose that $\phi(a)$ is true under v for every paramter a. By B_1, it follows that $\sim\phi(a)$ is false under v for every parameter a, meaning, by (2), that $\sim\forall x\phi(x)$ is not true, and is therefore false (because every sentence X is either assigned the value t or f), meaning, by B_1, $\forall x\phi(x)$ is true.

Case 2: In this case, γ is of the form $\sim\exists x\phi(x)$. So suppose that $\sim\phi(a)$ is true for every parameter a, meaning, by B_1, that $\phi(a)$ is false for every parameter a. Thus, by (2), $\exists x\phi(x)$ is not true, and is therefore false (for the same reason as in Case 1), meaning, by B_1, that $\sim\exists x\phi(x)$ is true.

(2): In this case, we need to prove that, if $\delta(a)$ is true for at least one parameter a under v, then δ is true under v.

Case 1: In this case, δ is of the form $\exists x\phi(x)$. So suppose that $\phi(a)$ is true for at least one parameter a, call it e. By B_1, $\sim\phi(e)$ is false, meaning, by (1), $\sim\exists x\phi(x)$ is not true, and is therefore false, and thus, by B_1, $\exists x\phi(x)$ is true.

Case 2: In this case, δ is of the form $\sim\forall x\phi(x)$. So suppose that $\sim\phi(a)$ is true for at least one parameter a, call it e. By B_1, $\phi(e)$ is false, meaning, by (1), $\forall x\phi(x)$ is not true, and therefore false, meaning, by B_1, $\sim\forall x\phi(x)$ is true.

2

We are to show that every Magic set is a truth-functional basis for propositional logic.

On the one hand, suppose that some pure sentence X is valid. Thus, it is true under all first-order valuations. We must show that X is a tautological consequence of some finite subset M_0 of M. That is, we must show that under every *Boolean* valuation that satisfies M_0, X is true. Well, by M_1, every Boolean valuation that satisfies S is also a first-order valuation. Thus, X is true under all Boolean valuations that satisfy M, because X was assumed to be true under all first-order valuations, and every Boolean valuation that satisfies M is also a first-order valuation. Thus, X is a tautological consequence of M. By the Corollary to the Denumerable Compactness Theorem for Propositional Logic, it is also a logical consequence of some finite subset M_0 of M.

On the other hand, suppose that some pure sentence X is a truth-functional consequence of some finite subset M_0 of M. Thus, $M_0 \bigcup \sim X$ is not first-order satisfiable (every first-order valuation is also a Boolean valuation). This means that $\sim X$ is not first-order satisfiable. For, if it were, by M_2, $M_0 \bigcup \sim X$ would also be first-order satisfiable. Thus, X is valid (true under every first-order valuation).

3

We must show that every complete regular set is also a magic set. Recall that a complete regular set M is a set such that for every γ, $\gamma \supset \gamma(a) \in M$ for every parameter a, and for every δ, $\delta \supset \delta(a) \in M$ for at least one parameter a, and that every finite subset of M is a regular set. Because we are showing that every complete regular set is a magic set, it suffices to show that M satisfies the conditions M_1, M_2.

M_1: We must show that every Boolean valuation that satisfies M is also a first-order valuation. To that end, it suffices to show that:
If γ is true, then $\gamma(a)$ is true for every parameter a,
If δ is true, then $\delta(a)$ is true for at least one parameter a.
(which implies a first-order valuation by Problem 1). So, suppose γ is true. Thus, $\gamma \supset \gamma(a) \in M$ for every parameter a. Because this is a Boolean valuation, and γ is true, $\gamma(a)$ is likewise true for $\gamma(a)$ for every parameter a (by B_4). On the other hand, suppose that δ is true. Thus, $\delta \supset \delta(a) \in M$ for at least one parameter a. Again, because this is a Boolean valuation, and δ is true, $\delta(a)$ is likewise true for at least one parameter a (again by B_4).

M_2: We must show that, for every finite set S_0 of pure sentences, and every finite subset M_0 of M, if S_0 is first-order satisfiable, so is $M_0 \bigcup S_0$. So, suppose that S_0 is a finite set of pure sentences, and that it is first-order satisfiable. Now consider any finite subset M_0 of M. By the definition of a complete regular set, it is a regular set. Thus, it can be split into a sequence

$$Q_1 \supset Q_1(a_1), ..., Q_n \supset Q_n(a_n)$$

where, if Q_i is a γ, a_i is any parameter a, and if Q_i is a δ, then a_i is a parameter a such that a does not occur in δ or in any previous Q's. So, what we really need to show is that, if a set S is first-order satisfiable, $S \bigcup \gamma \supset \gamma(a)$ is first-order satisfiable for every parameter a, and $S \bigcup \delta \supset \delta(a)$ is first-order satisfiable, provided a does not occur in δ or S. This would let us show that

$$S_0 \bigcup Q_1 \supset Q_1(a_1) \bigcup ... \bigcup Q_n \supset Q_n(a_n)$$

is first-order satisfiable, which is the same as saying $S_0 \bigcup M_0$ is first-order satisfiable. There are then two cases: Q_i is a γ, or is a δ.

γ: in this case, Q_i is a γ, meaning $Q_i \supset Q_i(a_i)$ is of the form $\gamma \supset \gamma(a_i)$. Suppose S is first-order satisfiable under some interpretation I. Well, $\gamma \supset \gamma(a_i)$ is satsified by *every* first-order valuation (which can easily be shown, if one desires), meaning $S \bigcup \gamma \supset \gamma(a_i)$ is obviously also first-order satisfiable.

δ: in this case, Q_i is a δ, meaning $Q_i \supset Q_i(a_i)$ is of the form $\delta \supset \delta(a_i)$. Suppose S is first-order satisfiable under some interpreation I. If δ is false under I, then $\delta \supset \delta(a_i)$ is true under I. If δ is true under I, then $S \bigcup \delta$ is satisfiable, meaning $S \bigcup \delta \supset \delta(a_i)$ is satisfiable by Problem 2 of Chapter 9 of the Beginners Guide[Smullyan, 2014] (because we assumed that a does not occur in δ or any of the previous $Q_i's$, and that S_0 was a set of pure sentences, meaning a of course does not occur in them). In either case, $S \bigcup \delta \supset \delta(a_i)$ is satsifiable.

4

We are to find a quicker proof of the Regularity theorem, using Magic sets. It has been shown that a complete regular set exists, and by Problem 3, it is a Magic set. By Problem 2, it is therefore a truth-functional basis for first-order logic. So, suppose that X is a valid pure sentence. Thus, X is truth-functionally implied by some finite subset M_0 of M. By the definition of a complete regular set, M_0 is a regular set. Thus, X is tautologically implied by some regular set, which is what we were required to prove.

System G_0

Correctness of G_0

The Axiom of the system is obviouslly correct. For, whenever the set of formulas U, X is true, X is true, meaning it is obviously true that *at least one* of the formulas V, X are true for any set of formulas V.

C_1: This is obvious, as X, Y are defined to both be true whenever $X \wedge Y$ is true.

C_2: If whenever all elements of U are true, at least one element of V, X and V, Y is true, on the one hand, if V is the element which is true, it follows that at least one element (V) of $V, X \wedge Y$ is true. On the other hand, if V is false, it would then follow that both X, Y are true, therefore meaning $X \wedge Y$ is the element which is true.

D_1: If whenever all elements of U are true then at least one element of V, X, Y is true, if V is the element which is true, it follows that at least one element of (V) of $V, X \vee Y$ is true. On the other hand, if V is false, it would then follow that either X or Y is true, therefore meaning at least one element of $V, X \vee Y$ is true.

D_2: We are given that at least one element of V is true whenever both U, X are true, or when both U, Y are true. By definition of $X \vee Y$, it follows that all elements of $U, X \vee Y$ are true iff either X or Y is true, which would imply that at least one element of V is true.

I_1: We are given that at least one element of V, Y is true whenever all elements of U, X are true. Thus, if all elements of U are true, either X is true or false. If it is false, at least one element $(X \supset Y)$ of $V, X \supset Y$ is true, while if it is true, it would follow by the assumptions that either V or Y is true. If it is V that is true, this is the element that is true. If it is Y that is true, then $X \supset Y$ is also true.

I_2: We are given that at least one element of V, X is true whenever U is true, and that at least one element of V is true whenever all elements of U, Y are true. Thus, if all elements of $U, X \supset Y$ are true, it follows that either V, X are true. If it is V that is true, we are done. If it is X that is true, we know that Y is also true, meaning V would also be true by the assumptions.

N_1: We are given that at least one element of V is true whenever all elements of U, X are true. If U is true, either X is true or false: if it is true, the assumptions prove that V is also true. If it is false, then $\sim X$ would be true by definition.

N_2: We are given that when all elements of U are true, at least one element of V, X is true. Thus, if all elements of $U, \sim X$ are true, V would be the true element implied by U, as X is false.

Problem 5

(a): On the one hand, suppose that the sequent $U, X \to V$ is true under some valuation v. There are two cases: either at least one of U, X are false under v, or both of U, X are true under v. In the first case, if it is U that is false, $U \to V, \sim X$ is automatically true. Otherwise, if X is false, $\sim X$ is true, meaning $U \to V, \sim X$ is also true. On the other hand, if both of U, X are true under v, V is also true under v, therefore meaning that $U \to V, \sim X$ is also true under v.

On the other hand, suppose that $U \to V, \sim X$ is true under some valu-

ation v. If U is false, we are done, as this would trivially mean that $U, X \to V$ is true. On the other hand, if U is true, either V or $\sim X$ is true as well. If it is V, we are done, as $U, X \to V$ would be true. If it is $\sim X$, then X would be false, therefore meaning $U, X \to V$ is trivially true.

(b): On the one hand, suppose that $U \to V, X$ is true. If U is true, either V or X is true. If it is X that is true, then $U, \sim X \to V$ is trivially true. If it is V that is true, then $U, \sim X \to V$ is of course also true. If U is false under v, then $U, \sim X \to V$ is trivially true under v.

On the other hand, suppose that $U, \sim X \to V$ is true under v. If both of $U, \sim X$ are true, then X is false, therefore meaning $U \to V, X$ is trivially true. On the other hand, either of $U, \sim X$ would be false. If it is U that it false, $U \to X, V$ is trivially true. If it is $\sim X$ that is false, X is true, therefore meaning $U \to V, X$ is also true under v.

6

As for the axiom, suppose that $|S, TX, FX|$ is provable, where S is a set of signed formulas. Let $TX_1, ..., TX_n$ be all of the signed formulas with a T in S, and $FY_1, ..., FY_k$ be all of the signed formulas with an F in S. Thus $|S, TX, FX|$ is just $X_1, ..., X_n, X \to X, Y_1, ..., Y_k$, which is just the same thing as $U, X \to V, X$ holding, as we could just let $X_1, ..., X_n$ be the set U, and $Y_1, ..., Y_k$ be the set V. On the other hand, suppose that we take the axiom of the system G_0. Thus, for any sets U, V of formulas, and any formula X, $U, X \to X, V$ is provable. Again, consider some set S of signed formulas, and suppose that $TX_1, ..., TX_n$ be all of the signed formulas with a T in S, and $FY_1, ..., FY_k$ be all of the signed formulas with an F in S. Thus $|S, TX, FX|$ is just $X_1, ..., X_n, X \to X, Y_1, ..., Y_k$. Because the axiom of G_0 holds for any sets U and V, we can simply take U to be the set $X_1, ..., X_n$, and V to be the set $Y_1, ..., Y_k$, thus proving that $|S, TX, FX|$ is provable.

We now need to show that the inferences rules of G_0 in uniform notation are equivalent to those inferences rules of G_0 not in uniform notation. First, let us show that the rules in uniform notation imply those rules not in uniform notation.

C_1: Consider any two sets of formulas U and V. We can take S to be the set of formulas $TX_1, ..., TX_n, FY_1, ..., FY_k$ such that $X_1, ..., X_n = U$ and $Y_1, ..., Y_k = V$ (for making the following proofs shorter, we will call the set S constructed in this way the *signed set for* U, V. By A, if $|S, TX, TY|$ is provable, that is, $U, X, Y \to V$, so is $|S, TX \wedge Y|$, that is, $U, X \wedge Y \to V$.

C_2: Consider the signed set S for any two sets U and V. By B, if $|S, FX|$ is provable, that is, $U \rightarrow V, X$, and $|S, FY|$ is provable, that is, $U \rightarrow V, Y$, then $|S, FX \wedge Y|$ is provable, that is, $U \rightarrow V, X \wedge Y$.

D_1: Consider the signed set S for U and V. By A, if $|S, FX, FY|$ is provable, that is, $U \rightarrow V, X, Y$, then $|S, FX \vee Y|$ is also provable, that is, $U \rightarrow V, X \vee Y$.

D_2: Consider the signed set S for U and V. By B, if $|S, TX|$ is provable, that is, $U, X \rightarrow V$, and $|S, TY|$, that is, $U, Y \rightarrow V$, is provable, then so is $|S, TX \vee Y|$, that is, $U, X \vee Y \rightarrow V$.

I_1: Consider the signed set S for U and V. By A, if $|S, TX, FY|$, that is, $U, X \rightarrow V, Y$, is provable, so is $|S, FX \supset Y|$, that is, $U \rightarrow V, X \supset Y$.

I_2: Consider the signed set S for U and V. By B, if $|S, FX|$, that is $U \rightarrow V, X$ is provable, and $|S, TY|$, that is $U, Y \rightarrow V$, is provable, so is $|S, TX \supset Y|$, that is $U, X \supset Y \rightarrow V$.

N_1: Consider the signed set S for U and V. By A, if $|S, TX|$, that is $U, X \rightarrow V$, is provable, so is $|S, F{\sim}X|$, that is $U \rightarrow V, {\sim}X$.

N_2: Consider the signed set S for U and V. By A, if $|S, FX|$, that is $U \rightarrow V, X$, is provable, so is $|S, T{\sim}X|$, that is $U, {\sim}X \rightarrow V$.

We now turn to showing that the inferences rules of G_0 imply A and B.

A: If α is $TX \wedge Y$, this can be shown using C_1. If α is $FX \vee Y$, this can be shown using D_1. If α is $FX \supset Y$, this can be shown using I_1. If α is $T{\sim}X$, this can be shown using N_2. If α is $F{\sim}X$, this can be shown using N_1.

B: If β is $TX \vee Y$, this can be shown using D_2. If β is $FX \wedge Y$, this can be shown using C_2. If β is $TX \supset Y$, this can be shown using I_2. The cases where β is $T{\sim}X$ or $F{\sim}X$ are the same as in the α case.

A set S is unsatisfiable iff its corresponding sequent $|S|$ is a tautology

On the one hand, suppose that a non-empty set S of signed formulas is unsatisfiable. There are then multiple cases. In the first case, suppose that there are no formulas of the form TX. Thus, $|S|$ is of the form $\rightarrow Y_1, ..., Y_k$ where every Y_i is the formula FY_i without the sign F. We must show that at least one of $Y_1, ..., Y_k$ is true. We are given that $FY_1, ..., FY_k$ is unsatisfiable, meaning, under

every interpretation I, there is some formula FY_j that is false. Under this same interpretation I, it is obvious that Y_j (removing the sign) is true, thus meaning at least one of $FY_1, ..., FY_k$ is true. In the second case, suppose that the set S contains no formulas of the form FY. Thus $|S|$ is of the form $X_1, ...X_n \rightarrow$. We need to show that at least one of $TX_1, ..., TX_n$ is false. But this is immediate, as it was assumed that one of $TX_1, ..., TX_n$ was false under every interpretation I, which is the same as saying one of $X_1, ..., X_n$ is false under every interpretation I. In the last case, the set S contains at least one formula of the form TX and at least one of the form FY. Thus $|S|$ is of the form $X_1, ..., X_n \rightarrow Y_1, ..., Y_k$. If every element of $X_1, ..., X_n$ is true, this would mean, by the assumption that S is not simultaneously satisfiable, that at least one of $FY_1, ..., FY_k$ is false (call this element FY_K). As before, it can be easily seen that Y_k would then be true under this same interpretation, therefore meaning at least one of $Y_1, ..., Y_k$ is true.

On the other hand, suppose that $|S|$ is a tautology for some non-empty set of formulas S. There are then 3 cases:

Case 1: In this case, S does not contain any formulas of the form TX. We know that $\rightarrow Y_1, ..., Y_k$ is a tautology, meaning under every interpretation, at least one of $Y_1, ..., Y_k$, which is the same as saying, under every interpreation, at least one of $FY_1, ..., FY_k$ is false, thus meaning S is unsatisfiable.

Case 2: In this case, S does not contain any formulas of the form FY. We know that $X_1, ..., X_n \rightarrow$ is a tautology, meaning, under every interpretation, at least one of $X_1, ..., X_n$ are false, which is the same as saying at least one of $TX_1, ..., TX_n$ are false, which means that S is unsatisfiable.

Case 3: In this case, S contains at least one formula of the form FY, and at least one formula of the form TX. Thus we know that $X_1, ..., X_n \rightarrow Y_1, ..., Y_K$ is a tautology. There are then two cases. In the first case, at least one of $X_1, ..., X_n$ are false under an interpretation I, which is the same as saying at least one of $TX_1, ..., TX_n$ is false under I, which would mean S is not simultaneously true under I. In the other case, all of $X_1, ..., X_n$ are true, which would mean that at least one of $Y_1, ..., Y_k$ are true, which is the same as saying that at least one of $FY_1, ..., FY_k$ is false, therefore meaning that S is also not simultaneously satisfiable under this interpretation.